AZITHROMYCIN

QT prolongation
Short 3–5 day course

CHEMICAL NAME (2R,3S,4R,5R,8R,10R,11R,12S,13S,14R)-11-[(2S,3R,4S,6R)-4-(dimethylamino)-3-hydroxy-6-methyloxan-2-yl]oxy-2-ethyl-3,4,10-trihydroxy-13-[(2R,4R,5S,6S)-5-hydroxy-4-methoxy-4,6-dimethyloxan-2-yl]oxy-3,5,6,8,10,12,14-heptamethyl-1-oxa-6-azacyclopentadecan-15-one

CLASS Macrolide (azalide) antibiotic

TRADE NAMES Zithromax · Azomax · Sumamed · Azithrocin

THERAPEUTIC USES / INDICATIONS

- Respiratory infections: community-acquired pneumonia (atypical cover), sinusitis, pharyngitis (penicillin allergy).
- Sexually transmitted infections: *Chlamydia* (single dose), adjunct for gonorrhoea.
- Typhoid fever; traveller's diarrhoea; trachoma.
- Prophylaxis of COPD exacerbations; *Mycobacterium avium* complex.

ADMINISTRATION

- **PO** tablets with or without food; **suspension** on empty stomach (product-dependent).
- **IV** infusion over ≥1 h (1–2 mg/mL) — not as bolus or IM.
- Separate from antacids containing Al/Mg by 2 h.

DOSAGES

- **Adult:** 500 mg once daily for 3 days, or 500 mg day 1 then 250 mg days 2–5.
- **Chlamydia:** 1 g single dose.
- **CAP (IV):** 500 mg once daily.
- **Children:** 10 mg/kg once daily for 3 days (max 500 mg/day).

PHARMACOKINETICS

- **Peak:** 2–3 h. High tissue and intracellular concentrations.
- Minimal metabolism; excreted mainly in bile.
- **t½:** 68 h → short courses give prolonged effect.

PHARMACODYNAMICS

- Binds **50S ribosomal subunit** → inhibits protein synthesis (bacteriostatic; bactericidal at high levels).
- Covers atypicals (*Mycoplasma*, *Chlamydia*, *Legionella*), *H. influenzae*, *M. catarrhalis*, streptococci.
- Anti-inflammatory effects in airways.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, abdominal pain.
- Headache, dizziness.
- Taste disturbance.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, torsades de pointes, sudden cardiac death.
- Hepatotoxicity.
- Hearing loss/tinnitus (high doses, long-term).
- *C. difficile* colitis; infantile hypertrophic pyloric stenosis (neonates).

CONTRAINDICATIONS & CAUTIONS

- Macrolide hypersensitivity.
- History of cholestatic jaundice with azithromycin.
- **Caution:** long QT, hypokalaemia, hypomagnesaemia, bradycardia, myasthenia gravis, liver disease.

KEY INTERACTIONS

- **Other QT-prolonging drugs** (amiodarone, haloperidol, ondansetron, fluoroquinolones): additive.
- **Warfarin:** ↑ INR.
- **Digoxin, colchicine:** ↑ levels (P-gp).
- **Antacids:** ↓ absorption.

NURSING MANAGEMENT

- Check ECG/QT and K^+, Mg^{2+} in at-risk patients.
- IV: infuse over at least 1 h; monitor site.
- Monitor LFTs in prolonged therapy.
- Assess respiratory symptoms and temperature.
- Report palpitations or fainting.

PATIENT EDUCATION

- Take the full short course; the drug keeps working after the last dose.
- Report palpitations, fainting, yellow skin.
- Partners need treatment for STIs.

KEY POINTS

- Macrolide; very long t½ → 3–5 day courses.
- QT prolongation risk.
- Covers atypical pneumonia and Chlamydia.
- Child 10 mg/kg once daily × 3 days.